Customer Shopping Behavior Analysis

Uncovering insights from **3,900 transactions** across product categories to guide strategic business decisions.

By NITISH GOPE

Dataset at a Glance

3,900

Purchases

Rows of transactional data

18

Features

Demographics, behavior & purchase details

37

Missing Values

In Review Rating column only

$59.76

Avg. Purchase

Average spend per transaction

Data Preparation in Python

01

Load & Explore

Imported with `pandas`; used `df.info()` and `.describe()`

02

Clean & Impute

Median imputation for missing Review Ratings by category

03

Feature Engineering

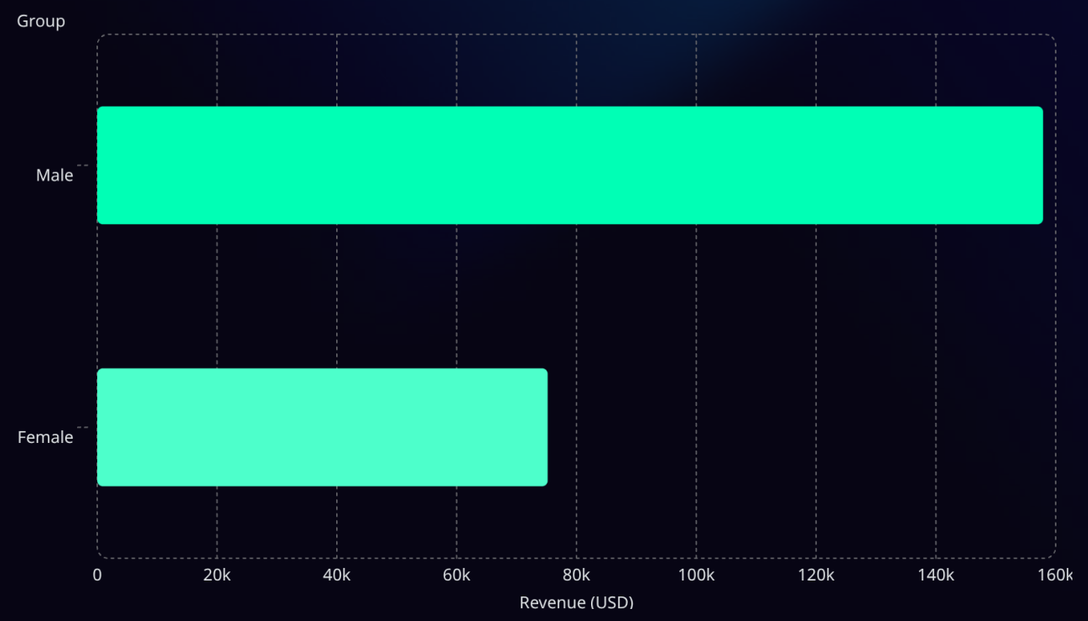
Created `age_group` and `purchase_frequency_days` columns

04

Load to PostgreSQL

Cleaned DataFrame pushed to database for SQL analysis

Revenue & Spending Insights



Key Findings

- Male customers generate **2× more revenue** than female
- **839 discount users** still spent above average purchase amount
- Express shipping users spend slightly more (**$60.48**) vs. Standard (**$58.46**)

Top Products & Discount Dependency

Top 5 by Rating

🥇 **Gloves**

3.86 avg rating

🥈 **Sandals**

3.84 avg rating

🥉 **Boots**

3.82 avg rating

Most Discount-Dependent

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation

Segment Breakdown

Loyal

3,116 customers — 5+ purchases

Returning

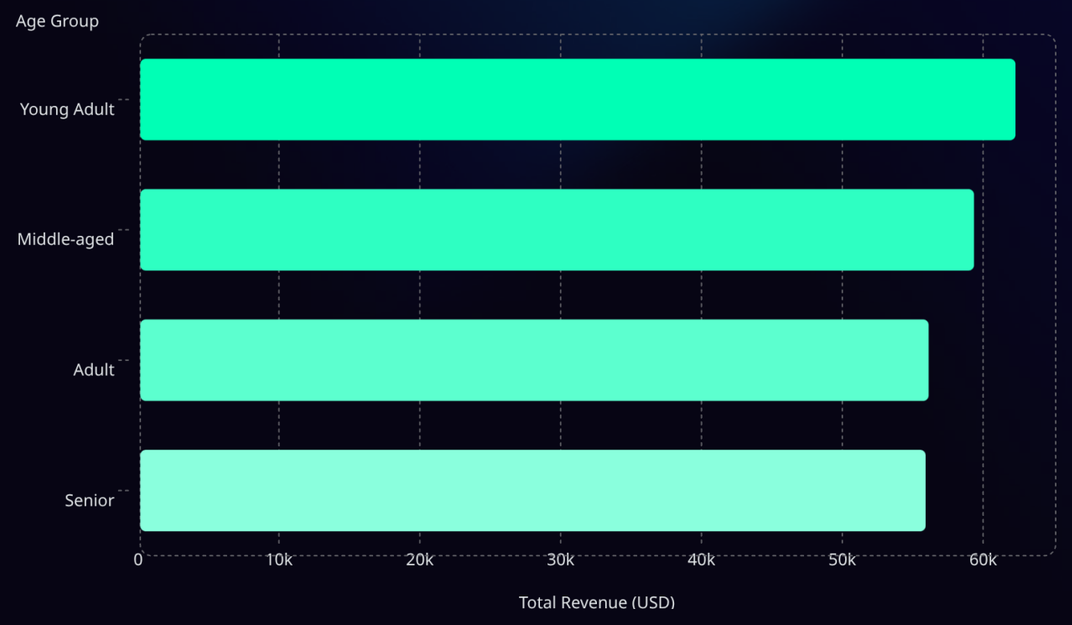
701 customers

New

83 customers

Repeat buyers with >5 purchases: **958 subscribed** vs. 2,518 not subscribed.

Subscriptions & Revenue by Age



Subscription Status

Status	Customers	Avg Spend	Total Revenue
Yes	1,053	$59.49	$62,645
No	2,847	$59.87	$170,436

Young Adults lead revenue at **$62,143**. Avg spend is nearly identical across subscription groups.

Top Products by Category

👗 Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

💍 Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

👟 Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

🧥 Outerwear

- Jacket — 163
- Coat — 161

Power BI Dashboard

3.9K Customers	**$59.76** Avg Purchase
3.75 Avg Rating	**27%** Subscribers

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to grow the 27% subscriber base

Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment

Review Discount Policy

Balance sales boosts with margin control on high-discount items

Targeted Marketing

Focus on Young Adults, top-rated products & express-shipping users